Portland International Airport (PDX)

Safety Management System (SMS) Manual

July 14, 2025

Port of Portland

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Introductory Statement

Airport Safety Management System (SMS) Manual
Portland International Airport (PDX)
Version 2025.1 – Effective 07/14/2025

This manual establishes the foundation for the PDX Safety Management System. It reflects our commitment to proactively manage safety, mitigate risks, and foster a safety culture that protects our employees, stakeholders, and the traveling public. As the designated Accountable Executive pursuant to 14 CFR §139.402(a)(1), I accept ultimate responsibility for ensuring the SMS is properly resourced, effectively implemented, and continually improved in accordance with applicable regulations and organizational goals.

I approve the contents of this Safety Management System Manual, which is promulgated pursuant to 14 CFR Part 139 Subpart E and follows the guidance set forth in the current edition of FAA Advisory Circular 150/5200-37. I affirm that the safety policies, processes, and procedures contained herein will be implemented and maintained as written.

This version of the SMS Manual supersedes all previous versions and will remain in effect until an updated version is formally revised.

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Dan Pippenger
Chief Aviation Officer
Portland International Airport

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Revision Date:______________________________

Record of Revisions

Date of Revision	Pages	Date of FAA Submission

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Revision Date:________________________________

Section 1 - Introduction to PDX Safety Management System

1.1 What is SMS?

A Safety Management System (SMS) is a structured approach to managing safety. It's an organized way of thinking about risk, decision-making, and safety accountability.

Organized management of safety brings clarity and consistency in how safety concerns are handled. The SMS provides a system for acknowledging and thoughtfully evaluating reported hazards. Those who submit concerns can track the status of their report, which builds confidence in the process. Over time, this transparency reinforces trust in the SMS and cultivates safety evolution within the community. The system is performance-based and built on four components: Safety Policy, Safety Risk Management, Safety Assurance, and Safety Promotion.

Beyond being a practical tool, SMS is a regulatory requirement. The PDX SMS is implemented in accordance with 14 CFR Part 139 Subpart E and follows the guidance set forth in the current edition of FAA Advisory Circular 150/5200-37. SMS will establish a unified and evidence-driven system for airports to anticipate and manage operational safety risks while ensuring safety responsibilities are defined and embedded throughout the organization.

1.2 Purpose

This manual describes the structure and operation of the PDX SMS and serves as PDX's primary documentation for demonstrating compliance with SMS requirements. It outlines the functions of the SMS program, roles and responsibilities within the SMS, and how risks are identified, evaluated and addressed. It also serves as the guiding document for staff and stakeholders participating in the SMS and is formally referenced in the PDX Airport Certification Manual (ACM) to ensure alignment with regulatory oversight.

1.3 Scope and Applicability

The SMS applies to everyone with access to the Air Operations Area (AOA) or whose activities may directly impact its safety. Participation in the SMS is not optional; all applicable parties are expected to engage in safety reporting, support mitigation efforts, comply with established procedures, and respond to safety communications in a timely and responsible manner. These expectations apply regardless of employer or contractual relationship and are essential to ensuring consistent, effective risk management across the airside environment. See Figure 1 for a map of the SMS area of oversight.

The scope of the SMS includes all operational activities, infrastructure, and procedures within the AOA, as well as any projects, changes, or events that could introduce new hazards or affect airside safety.

Activities include, but are not limited to, airfield inspections, aircraft operations, vehicle operations, maintenance or construction within or adjacent to the AOA, wildlife management, fueling, and emergency response.

While the SMS covers a wide range of operational activities, it does not replace those functions. Port of Portland employees, airport tenants, contractors, and others with operational responsibility will continue to lead and manage their respective areas. The goal is to ensure consistent safety accountability in all airside operations. Safety concerns, observed hazards, and airport operational changes must be reported or captured in the system. This may occur through direct hazard reporting, communication of planned physical or procedural changes, or through data collected from existing systems.

Even when a reported hazard or change doesn't result in a risk assessment, the data that is captured plays a crucial role in the safety picture. The SMS uses this data to identify patterns, track repeated conditions, and spot areas where underlying risks may be going unnoticed or unaddressed.

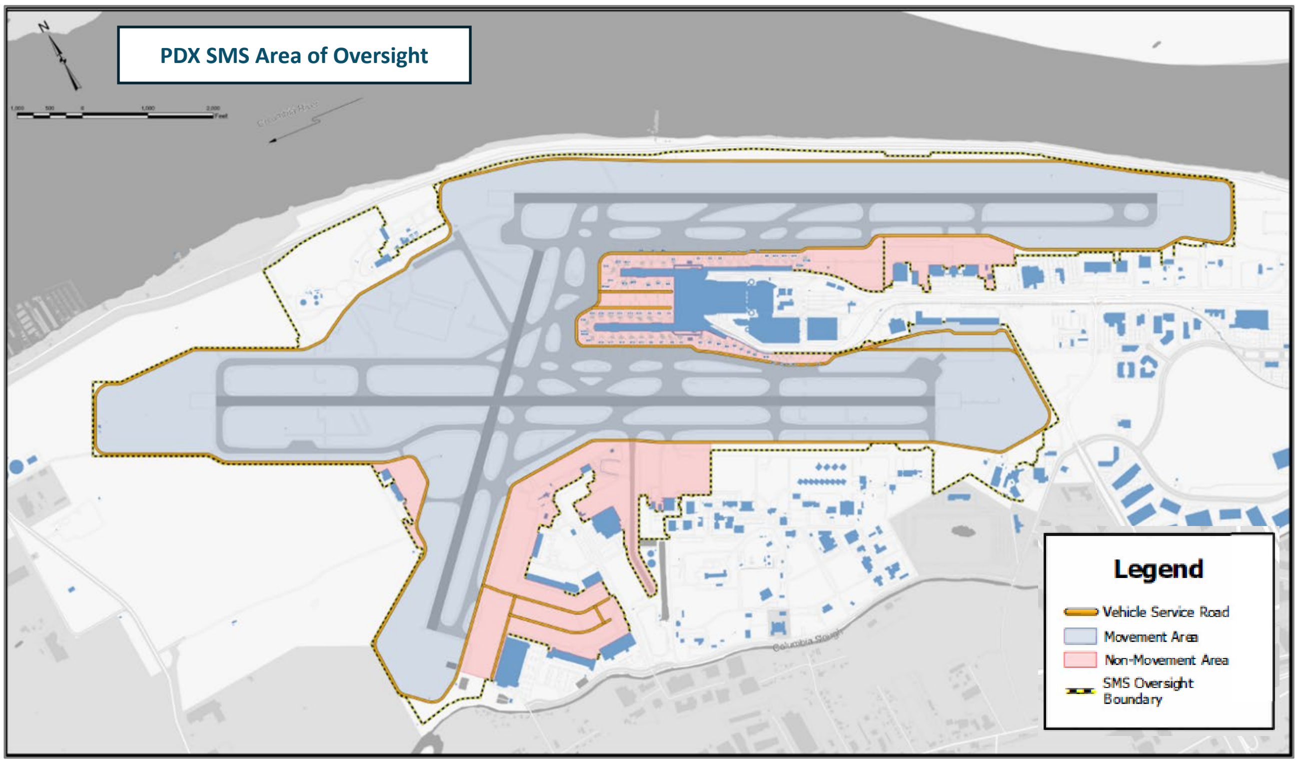

Figure 1 SMS Oversight Map. Any operation occurring within the AOA perimeter is part of the PDX Safety Management System. Note that the bag tunnel is included in the SMS area of oversight. Click the thumbnail to view the zoomable map in Appendix C.

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1.4 Document Access

The PDX SMS Manual is readily available to all stakeholders operating within the AOA, ensuring that everyone has access to the latest procedures and guidelines. The manual is available to PDX badgeholders on flypdx.com/SMS.

1.5 Manual Review and Revisions

The SMS Manual will be reviewed annually to ensure it remains current, functional, and aligned with the requirements of 14 CFR Part 139 Subpart E. The SMS Team will coordinate the annual review process, which includes gathering input from participating departments, subject matter experts, and the SMS Executive Committee. The results of this review will be formally submitted to the Accountable Executive (AE) for approval.

Revisions can be made at any time, if necessary, to reflect significant operational changes, regulatory updates, process improvements, or corrective actions. Requests for changes to the manual can be submitted by any airport department or stakeholder participating in the SMS. All revision requests should be submitted in writing to the SMS Team (SMS@flypdx.com) and include the following information:

1. Name
2. Company
3. Contact information
4. Rationale for the change
5. Supporting documentation or regulatory references
6. Description of the expected impact

Proposed changes will be evaluated by the SMS Team for relevance, accuracy, and alignment with regulatory and operational priorities. If deemed appropriate, the proposed revision will be included for consideration during the next annual review cycle. All approved revisions will be recorded in the revision history log and distributed per the Document Access section.

In accordance with 14 CFR § 139.401(g), on an annual basis or upon FAA request, PDX will provide the FAA with a list of all changes made to this SMS Manual. This requirement will typically be fulfilled during the annual Part 139 certification inspection. Any urgent or significant changes will be communicated outside of this cycle, as appropriate, to maintain compliance and ensure ongoing alignment with regulatory expectations.

Section 2 - Safety Policy

2.1 Introduction

The PDX SMS is rooted in a culture of proactive, accountable safety leadership. The Safety Policy defines the values and expectations that inform SMS components and formalizes the obligation of airport leadership and staff to participate in and support the SMS.

2.2 Safety Policy Statement

The Safety Policy Statement is endorsed by the Accountable Executive (AE), the Responsible Executive (RE), and the Responsible Manager (RM). It serves as a visible demonstration of PDX senior leadership's safety accountability and defines PDX's safety vision and values, dedication to a just safety culture, and commitment to continual improvement. A copy of the signed Safety Policy Statement is available online (flypdx.com/SMS) to all airport personnel and tenants to ensure transparency, accessibility, and shared understanding of PDX's safety commitments.

View the PDX SMS Safety Policy Statement in Appendix B

The following sections operationalize the Safety Policy Statement, detailing the expectations and structures that enable PDX to function as a high-performing, forward-leaning safety organization.

2.3 Just Safety Culture

The PDX SMS is founded on the principles of a *just safety culture*, which balances accountability with learning. A just culture recognizes that while individuals are responsible for their actions, most errors are unintentional and best addressed through systems improvement and education, not punishment. Just culture is essential to building trust, maintaining transparency, and enabling the continual improvement of airport safety.

Under a just culture, employees are encouraged to report safety concerns, hazards, and incidents without fear of reprisal, provided their actions were not reckless or intentionally harmful. This approach ensures that safety data is not suppressed and that organizational learning is prioritized.

PDX focuses on identifying root causes, improving systems, and supporting staff in maintaining safe operations.

2.4 Safety Objectives

The PDX SMS is designed to improve operational safety performance through proactive hazard identification, consistent risk assessment, and informed decision-making. PDX is committed to establishing measurable safety objectives that align with both regulatory expectations and operational realities.

PDX is in the early stages of SMS implementation and does not yet have the volume or quality of data necessary to define meaningful, data-driven safety objectives. Rather than include placeholder objectives that may not reflect true operational needs or risk profiles, PDX will first collect and analyze baseline safety performance data. This includes hazard reporting and location trends, safety investigation findings, risk assessment outcomes, incident data, and mitigation timelines.

During the implementation phase, the SMS Manager will evaluate this data to inform the development of objectives that are specific, measurable, achievable, relevant, and time-bound (SMART). These objectives will be incorporated into the Manual following approval of the SMS Executive Committee.

Once established, safety objectives will be reviewed and monitored as part of Safety Assurance activities. Objectives will emphasize measurable improvements in key areas such as hazard reporting, mitigation effectiveness, and airport-wide safety collaboration.

The processes described throughout this manual form the operational framework by which PDX will meet its established safety objectives. The SMS organizational structure is designed to ensure that each function has a defined role in achieving these objectives through data-driven decision-making.

2.5 Organizational Structure

2.5.1 Roles and Responsibilities

The following roles comprise our internal safety organization. Roles beyond these (including tenants and other external stakeholders) may participate through committees or other operational interfaces.

- **Accountable Executive (AE):** Ultimately responsible for SMS performance, resource allocation, and ensuring that all SMS activities comply with requirements in 14 CFR Part 139 Subpart E. The AE receives periodic and urgent reports regarding critical risks and compliance issues. Reviews and approves mitigation strategies and/or risk acceptance for high risks and interim critical or escalated risks. The Accountable Executive role is filled by the Chief Aviation Officer for the Port of Portland.

- **Responsible Executive (RE):** Provides oversight of safety initiatives, monitors system performance, and works with the AE to implement strategic improvements. Reviews and approves mitigation strategies and/or risk acceptance for high or escalated risks. The Responsible Executive role is filled by the Director, Airport Operations.

- **Responsible Manager (RM):** Oversees daily SMS operations at the operational level, ensuring hazard reports, risk assessments, and corrective actions are executed effectively. Reviews and approves mitigation strategies and/or risk acceptance for medium and low risks. The Responsible Manager role is filled by the Sr. Manager, PDX Airside Operations.

- **SMS Manager:** Manages the day-to-day SMS processes, including administering the hazard reporting system, coordinating risk assessments, and ensuring documentation and follow-up. The SMS Manager is also responsible for coordinating safety investigations when incidents, near misses, or significant safety concerns occur within the AOA. While operations, public safety and security, or other teams may conduct initial incident response and reporting and/or additional incident follow up for other purposes, the SMS investigation focuses on identifying contributing factors, potential systemic issues, and opportunities for improvement. Investigations may involve subject matter experts and findings will be documented and reviewed by the SMS Rapid Review Team or escalated to the Airside Safety Committee.

- **SMS Coordinator:** Supports the SMS Manager in data collection and trend analysis, report preparation, safety investigations, safety promotion and training, risk assessment and committee facilitation, and hazard identification. Leads targeted safety efforts as needed.
- **SMS Team:** Led by the SMS Manager, the SMS Team advocates and promotes the SMS. Team members represent diverse subject matter expertise and a cross-section of airport work groups. The SMS Team may assist in the review of hazards and evaluation of risks, coordinate safety promotion materials, and conduct safety assurance activities.

2.5.2 SMS Committees and Teams

The PDX SMS is supported by a structured set of committees and teams that provide oversight, review, and opportunities for collaborative engagement among AOA stakeholders. Each group plays a unique role in ensuring safety risks are evaluated, mitigations are vetted, and the system functions as intended.

SMS meetings listed in this section will be documented through meeting minutes or action logs. These records will include attendee names, items discussed, decisions made, action items assigned, and any risk assessments initiated or closed. Committee meeting minutes will be reviewed and approved by the committee chair or facilitator and retained in accordance with applicable federal and state records retention requirements.

SMS Rapid Review Team

Purpose: To perform regular reviews of newly reported hazards, determine appropriate next steps (closure, escalation, or referral), and complete preliminary hazard assessments (PHA). This team ensures that each report is triaged promptly and enters the appropriate risk review pathway.

Facilitator: SMS Manager or delegate

Typical Meeting Cadence: Biweekly. Ad hoc sessions convened as needed.

Participants: Five core voting members drawn from key Port of Portland departments. These may include: Airport Operations, Occupational Safety, Public Safety & Security, Construction, and Airport Business & Properties. Additional subject matter experts (SMEs) may be invited to specific meetings based on the nature of the hazard, including, but not limited to:

Maintenance
Wildlife
Environmental
Airport Communications Center
Airside Planning
Landside/Terminal (for overlapping impacts)
Fire/EMS
Port Police
Emergency Management
Aviation Security
Central Projects Office
Aviation Long-Range Planning
Legal
Construction (tenant or capital improvement)

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Public Affairs
UAS Program
Engineering
Information Technology

SMS Airside Safety Committee

Purpose: To collaboratively review hazards and mitigations that require broader operational input, particularly those involving shared responsibilities or external stakeholders. The committee also serves as a regular forum for discussing ongoing safety issues, trends, and improvements in the airside environment.

Chair: SMS Manager or delegate

Typical Meeting Cadence: Monthly. Additional meetings convened as needed.

Participants may include: Internal staff from airside-related departments (Operations, Maintenance, Safety, Environmental, Aviation Security, etc.), and external stakeholders including airline representatives, ground handlers, and others operating in the AOA. The first portion of each meeting may be conducted with internal Port participants only, with the second portion opened to external attendees.

SMS Executive Committee

Purpose: The SMS Executive Committee serves as the senior decision-making body for PDX SMS oversight. This group is responsible for reviewing critical risks, validating the effectiveness of risk mitigations, and approving changes to the SMS program and manual. The committee's purpose is to take informed, outcome-driven action that strengthens PDX's safety performance. Meetings focus on meaningful data, unresolved risks, cross-departmental barriers, and strategic decisions that require executive alignment or intervention.

Chair: Accountable Executive or delegate

Typical Meeting Cadence: Annually. Additional meetings may be convened if triggered by critical risk reports, regulatory changes, major safety events, or by SMS leadership request.

Participants may include: Accountable Executive, Responsible Executive, Responsible Manager, SMS Manager, senior leaders from Operations, Public Safety & Security, Environmental, Legal, Risk, Occupational Safety, Public Affairs, Aviation Business & Properties, and others.

PDX SMS Organization Chart

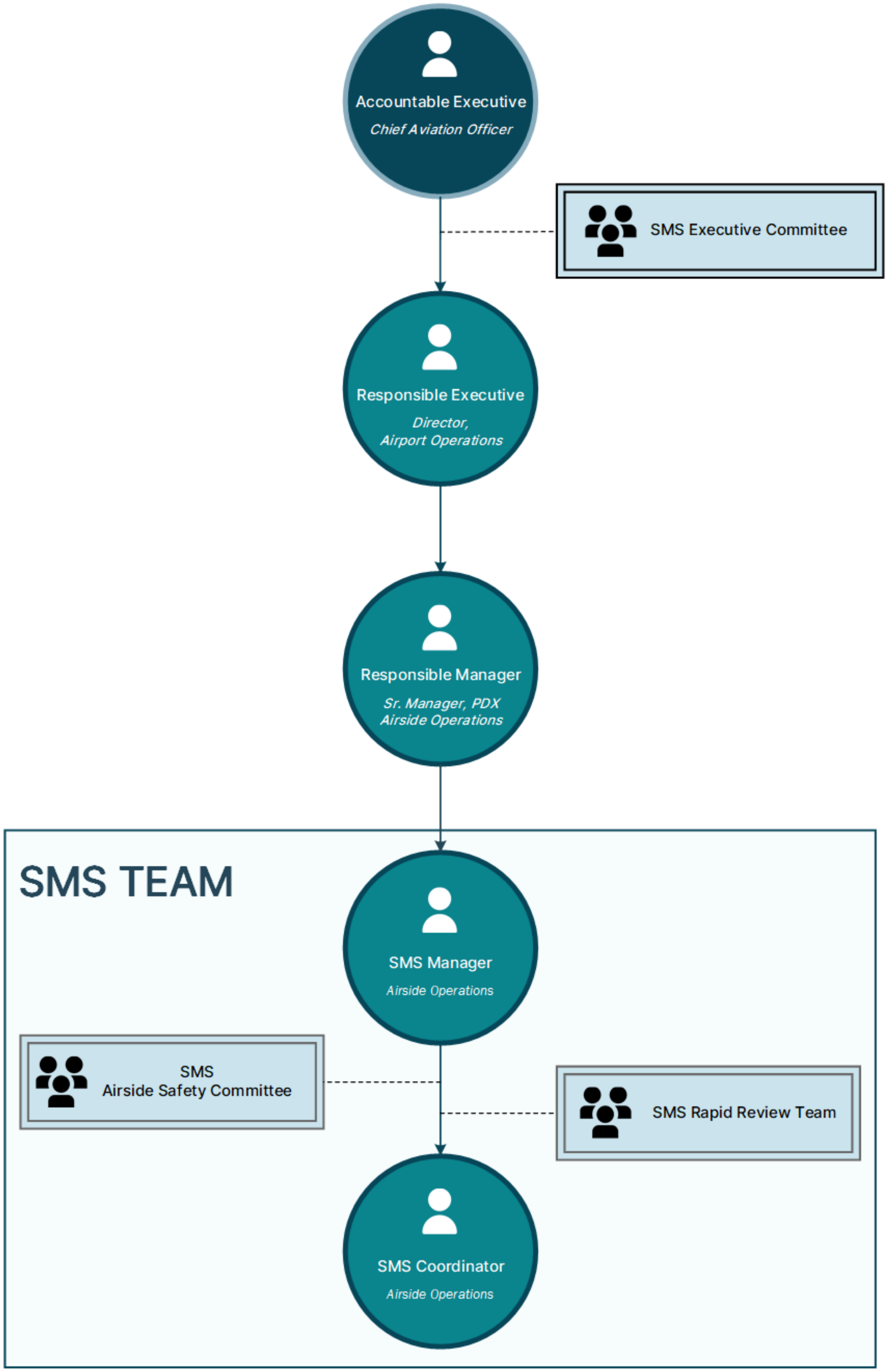

Figure 2 PDX SMS Organization Chart

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Section 3 - Safety Risk Management

3.1 Introduction

Safety Risk Management (SRM) is the structured process through which PDX identifies hazards, analyzes and assesses associated risks, and determines appropriate mitigation strategies. It is essential to ensuring operational safety risks are understood, addressed, and reduced to acceptable levels.

The SRM process at PDX is designed to be practical, transparent, and scalable. It ensures that every reported hazard is reviewed, categorized, and either closed, delegated, or escalated based on the potential risk. This process is supported by departmental subject matter experts, frontline employees, and the leadership structure defined in this manual.

PDX's approach to SRM balances regulatory compliance with operational practicality. Risk-based decision making ensures that appropriate resources and subject matter expertise are applied to each routine or high-consequence situation. This process promotes consistency in decision-making and establishes a clear record of how safety risks are managed across the organization.

3.2 Hazard Reporting

Hazards may be identified through a variety of channels, including employee observation and reporting, operational oversight, data monitoring, or during the planning or execution of operational or procedural change. The SMS is structured to ensure these observations are captured through an integrated, cloud-based reporting and intake process.

All personnel working within the AOA are expected to report any unsafe condition, process, or practice. Any change where airside infrastructure, polices, or procedures are developed or revised shall be reported to the SMS Team at inception.

Hazard Identification

Hazard reports can be submitted through the PDX hazard reporting system. This tool enables intake, triage, and tracking of hazards. The system also allows reporters to follow the status of submitted concerns via a unique tracking ID. The system is configured to route entries to the SMS Team for review and action.

Hazards may also be identified through data trending related to key airside safety functions, including, but not limited to: Airside Operations, Maintenance, Wildlife, Fire/EMS, Environmental, Construction and Planning, Emergency Management, Risk, Port PD, Comm Center, and others.

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3.2.1 Reporter confidentiality

The success of the SMS depends on broad and open participation, and that participation is only possible when individuals feel safe, supported, and heard. By encouraging the reporting of hazards, PDX fosters a culture of shared responsibility where safety is continuously improved through collaboration and transparency. All personnel are encouraged to report hazards and safety concerns without fear of reprisal.

Reports may be submitted anonymously. Identifying information can facilitate reporting accuracy and support safety investigations, but it is not required.

Consistent with a just culture, a system will also be developed that provides a means for reporting confidentially (to the extent permissible by law) and that identifies applicable exceptions. Exceptions to this confidentiality protection may include, for example, intentional falsification of safety reports or knowingly neglecting to report safety-related issues.

3.3 How Hazards are Processed

The SRM process begins with the identification of a hazard or operational change that could impact safety. Upon receipt, the SMS Team screens the information for completeness, validates the hazard, and categorizes the report as either a new hazard, a potential change, or if it does not fall within the SMS scope, it is closed or referred elsewhere. Reported hazards that have previously been evaluated will be captured as a data point and may be re-assessed if the data indicates a change necessitating additional review.

Valid safety concerns are forwarded to the SMS Rapid Review Team (RRT) for triage and Preliminary Hazard Assessment (PHA) processing. The RRT will determine whether the hazard should be closed, referred, or escalated for further assessment based on the complexity or PHA. The hazard can be escalated to a Limited Risk Assessment (LRA), led by a group of identified SMEs or members of the Airside Safety Committee, or a Technical Risk Assessment (TRA) can be initiated (see Table 2). For hazards scored at high or above during the PHA, a TRA shall be conducted. The SMS Team may also initiate a TRA at their discretion. Potential TRA triggers are listed in Table 1.

PDX SMS POTENTIAL TECHNICAL RISK ASSESSMENT TRIGGERS	
Trigger	**Examples (not exhaustive)**
Operational Changes	Introduction of new or changing equipment, technology, or aircraft; changes to taxi procedures; traffic flow on VSR, updated operations SOPs, etc.
Projects, Construction, Infrastructure changes	Passenger Boarding Bridge (PBB) replacement, electric Ground Service Equipment (eGSE) charging station installation, changes to airfield layout, construction in bag tunnel, etc.
Incidents, Accidents, Close Calls	Large fuel spill, aircraft or ground vehicle collision, runway incursion or excursion, etc.
New or Changing Regulations or PDX Rules	PDX rules impacting any airside operation, updates to Part 139, Part 77, Part 107, new CertAlerts, etc.

Safety Events w/Escalating or Repetitive Trends	Persistent Part 139 discrepancies, excessive time to resolution thresholds for discrepancies, repeated reports of safety hazards in bag tunnel hot spots, etc.
High or Critical Initial Risk Ratings	During a preliminary hazard assessment, a hazard is scored as high or critical risk
Emerging or Unfamiliar Hazards	New technologies, operations, or tenants
Complex or Cross-Departmental / Multi-Stakeholder Impacts	AOA closures, openings, alleyway taxilane changes, etc.

Table 1 – Potential Technical Risk Assessment (TRA) triggers

Table 2 details the various levels of analysis a hazard may go through, based on the level of complexity.

PDX SMS HAZARD ASSESSMENTS			
Assessment Level	**Purpose**	**Conducted By**	**Characteristics**
Preliminary Hazard Assessment (PHA)	Initial hazard identification and screening	SMS Manager, SMS Coordinator or SMS Rapid Review Team	Rapid screening and triage of hazards to determine need for further analysis
Limited Risk Assessment (LRA)	Assess operational hazards requiring more than preliminary review	Operational staff with safety oversight, SMEs as needed, or SMS Airside Safety Committee	Focus on day-to-day risks, process hazards, informal setting
Technical Risk Assessment (TRA)	Comprehensive evaluation of significant hazards	Multidisciplinary panels	Detailed analysis, required development of mitigation strategies

Table 2 Hazard Assessments

Whether the hazard only goes through a PHA or is evaluated through a LRA or TRA, PDX will use the five-step risk assessment process to ensure consistent, structured, and data-informed analysis.

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3.3.1 Five-Step Risk Assessment Process

The five-step risk assessment process evaluates hazards, determines the level of associated risk, and identifies appropriate mitigations (if applicable). This standardized approach ensures that safety decisions are based on credible data, stakeholder input, and consistent evaluation criteria. Whether applied to a newly identified hazard or a planned operational change, the five-step process supports proactive safety management and informed risk management at all levels of the organization.

The following section presents a high-level overview of each step in the risk assessment process, providing the foundational understanding needed for day-to-day application.

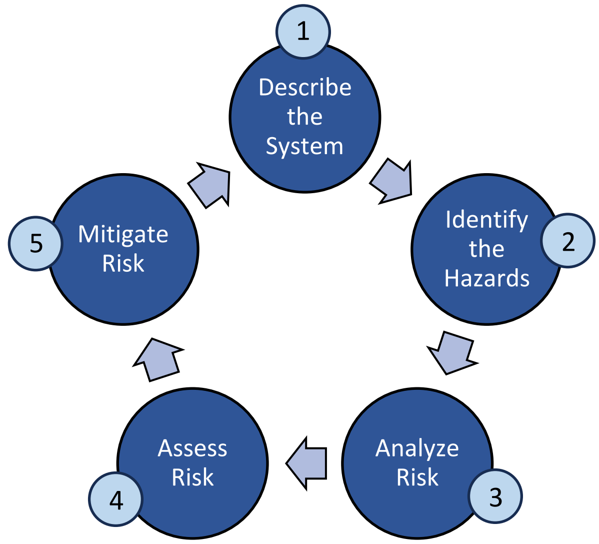

Figure 3. The Five-Step Risk Assessment Process

Step 1: Describe the System

Each assessment begins by clearly defining the system (operational environment and operational context) in which the hazard or change exists. PDX utilizes the 5M Model, which provides a structure for defining how different operational factors interact to influence safety outcomes.

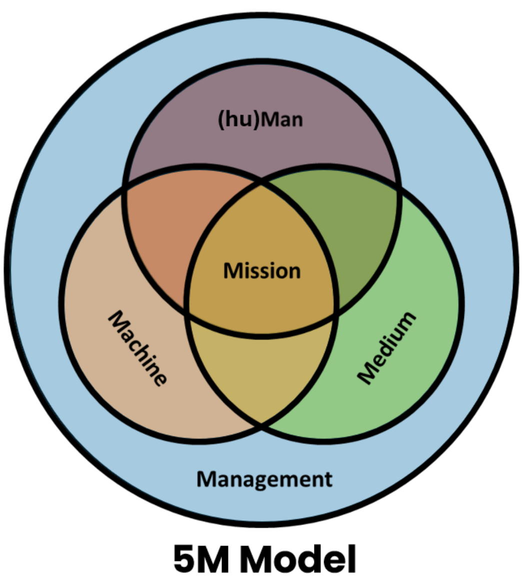

Figure 4. The 5M Model is used in Safety Risk Management to categorize a hazard's operational environmental context

1. **(hu)Man/People**: The personnel involved (e.g., airside operations, maintenance, ARFF, tenants, ground handlers)
2. **Machine**: The equipment, vehicles, and technologies used and their interactions (e.g., hardware, firmware, software, human-to-system interface, avionics, etc.)
3. **Medium**: The physical environment (e.g., airfield layout, construction zones, ramp areas, weather, wildlife activity, etc.)
4. **Management**: The operations, processes, and procedures in place, including current or expected changes
5. **Mission**: type of operation, objective of the operation, operational specifics and complexities

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Step 2. Identify the Hazards

Once the system is described, the team identifies the specific hazard(s). A hazard is defined as a condition, object, or activity that has the potential to cause injury, damage, or disruption.

Hazards can include, but are not limited to:

- Unsafe conditions (e.g., deteriorating markings, low lighting)
- Unsafe acts (e.g., improper vehicle operation)
- Design flaws or procedural gaps (e.g., jet blast exposure or no standard process for escorting contractors in the movement area)
- External/environmental factors (e.g., wildlife activity, unusual weather conditions)
- Operational changes (e.g., new taxi procedures, changes in traffic flow, changes in ramp equipment)

Step 3: Analyze the Risk

For each hazard, the team will determine the worst credible outcome. This means imagining the most severe adverse result that is realistic under the circumstances, not an outlandish worst-case that defies reality. To determine the worst credible outcome, consider the current operating context and controls in place, then ask: If this hazard leads to an incident, what is the worst harm we can reasonably expect?

The outcome should be supported by data or expert experience as a plausible scenario. For example, if analyzing a fuel spill hazard, the worst credible outcome might be a fire causing severe injuries and damage, if such a scenario is plausible given fuel quantities and ignition sources. It would not stretch to an impossible scenario (like the spill causing an entire airport-wide disaster) if that isn't credible. By concentrating on credible outcomes, exaggerating or trivializing hazards is avoided.

Step 4: Assess the Risk

For each hazard, the team evaluates the potential consequence and likelihood of occurrence to determine the initial risk level. Severity is assessed based on the worst credible outcome, considering potential harm to people, equipment, operations, and reputation. Likelihood is assessed based on frequency occurrence, PDX and industry historical data, SME input, and system complexity.

PDX uses a 5x5 risk matrix to determine the risk. The PDX Risk Matrix and Risk Acceptance Authority Table can be found in Appendix D. Each level of severity and likelihood is defined and supported by real-world examples and operational context. The resulting risk level is categorized as Low, Medium, High, or Critical.

With the initial risk level established, the rating will determine:

- Whether the risk is acceptable without further action
- Whether mitigation is recommended or required
- The level of leadership approval required

Step 5: Mitigate the Risk

If the risk is not acceptable or mitigations are recommended, the team develops one or more mitigation strategies, which may include:

- Preventative actions (e.g., new SOPs, additional training, refreshed markings)
- Physical modifications (e.g., lighting, barriers, signage)
- Administrative changes (e.g., access limitations, schedule revision, additional staff)
- Technological solutions (e.g., radar, alerts, automation, software)

Mitigation Strategies are evaluated using the PDX Risk Matrix to identify the residual risk. If the residual risk remains outside of acceptable parameters, the mitigation plan is revised until the risk is acceptable.

At PDX, mitigations are tracked through the SMS hazard management system using timelines, assigned ownership, and documentation. To ensure proper documentation and follow up, each mitigation plan should identify the following:

- Responsible department or role (and primary POC)
- Implementation steps and timeline
- Monitoring plan (audits, inspections, data trend analysis)
- Expected outcome (what does success look like?)
- Residual risk score based on updated likelihood and severity

Mitigations are documented in the SMS hazard tracking system and remain open until verified as complete and effective through the safety assurance process.

The results of risk assessments inform subsequent training, system adjustments, and process refinement, ensuring that our SMS evolves as operational conditions change.

3.3.2 Risk Mitigation Tracking and Residual Risk Verification

Once mitigation strategies are implemented, the final responsibility of the SRM process is to verify the effectiveness and confirm that residual risk has been reduced.

The SMS Team monitors mitigation implementation progress and evaluates whether mitigations have achieved the desired reduction in risk. Incomplete, ineffective, or delayed mitigations may result in reassessment or escalation to senior leadership.

Residual risk is monitored and recalculated (if needed, based on new or additional information) using the same risk matrix applied during the initial analysis. If the residual risk remains outside of acceptable parameters, the mitigation plan is re-evaluated and revised until the risk is acceptable.

3.3.3 Records Retention

SMS-related records will be retained in accordance with applicable law, including 14 CFR Part 139.

Section 4 - Safety Assurance

4.1 Introduction

Safety Assurance (SA) is responsible for monitoring safety performance, verifying the effectiveness of risk mitigations, identifying system deficiencies, and supporting continual safety improvements. It ensures that hazard risk controls remain effective over time.

This is achieved through a structured set of internal reviews, data analysis, and feedback processes that enable PDX to detect emerging risks, validate that safety actions are working as intended, and adapt to a changing operational environment.

4.1.1 Data Collection and Integration

PDX leverages business intelligence (BI) tools to support SA activities. The SMS platform will integrate with multiple data sources and feed into a BI tool for data visualization, trending, analysis, and strategic insight.

Data sources utilized for SMS may include:

- Hazards submitted through the SMS hazard reporting system
- Direct communications with the SMS Team
- Incidents, near-misses, citations, and inspection data
- Maintenance work order data
- Access control data
- Wildlife and environmental activity logs
- Data from airline partners, tenants, and ground handlers

These data sources may be routinely ingested into a centralized BI dashboard that consolidates, organizes, and filters records by type, frequency, location, and other parameters.

4.1.2 Review and Validation

The SMS Team conducts targeted inspections to confirm that mitigations have been implemented as intended and engages with departments to ensure updated procedures are understood and consistently applied. Internal audits evaluate whether reports are being triaged appropriately, whether risk acceptance decisions are being made at the proper level of authority, and whether the system itself is functioning within the expectations set forth in this manual.

After mitigation implementation, SA maintains oversight of those mitigations to confirm they are functioning as intended. If a control measure fails to produce the expected outcome, the SMS Team may require a reassessment.

4.1.3 Assurance Activities

- **Regular Inspections and Audits**: Targeted internal audits may be conducted on select SMS processes to assess whether the system is functioning as intended. Audits are performed annually or as needed.
- **Monitoring of Safety Performance Indicators (SPIs):** Key metrics are tracked to identify trends and systemic issues, which may include:
  - Hazard report submission rates
  - Resolution times
  - Percentage of mitigations implemented within planned timeline
  - Number of repeat safety findings from audits or inspections
  - Percentage of corrective actions closed on time
- **Trend Analysis**: Monitored data are analyzed to detect patterns, informing training needs, process improvements, and hazard clusters.
- **Leadership Reviews:** The Accountable Executive and the SMS Executive Committee will review audit findings and performance data, allowing them to make strategic decisions on resource allocation and system changes.
- **Corrective Action Tracking:** Any non-compliance or suboptimal procedures trigger corrective action plans. These are documented and monitored to closure. Corrective Action Plans (CAPs) are developed when a problem is found in the system in order to fix a deficiency or nonconformance with existing systems, mitigations, or processes. CAPs are often triggered after a failure, noncompliance, or ineffective mitigation is discovered.

4.1.4 Data Transparency and Safety Assurance Feedback

Findings from SA activities are routinely reviewed by the SMS Executive Committee and shared with operational leadership. Emerging trends, recurring issues, and audit results are summarized in the Safety Assurance Report and used to inform decisions about training, procedures, staffing, and technology investment.

Information obtained through SA activities will be routinely shared with AOA stakeholders to support transparency, promote shared responsibility, and strengthen safety awareness across the airport environment. This communication may take the form of safety bulletins, summary reports, dashboards, trainings, email alerts, or presentations during stakeholder briefings and committee meetings. By sharing relevant insights in a timely and accessible way, PDX ensures that those working in the AOA are informed and engaged. The process of communicating safety-related information is provided in further detail in the Safety Promotion section.

Safety Assurance also plays a role in confirming whether PDX is making measurable progress toward its safety objectives. Safety Performance Indicators (SPIs) are established as part of the development of PDX's safety objectives. The SPIs are tracked over time and are reviewed by the SMS Manager and SMS Executive Committee. These indicators serve as a feedback mechanism to determine whether mitigation strategies, training programs, reporting systems, and safety culture initiatives are delivering their intended outcomes. If SPIs indicate stagnation, decline, or unexpected results, the SMS Team may require an internal review, revise the objective(s), or initiate a new risk assessment.

Safety Assurance is an active, adaptive function that allows PDX to learn from its operations, validate its safety strategies, and remain prepared for the complexities of growth and change. It ensures that the SMS is an evolving system capable of maintaining safety in a dynamic, high-risk environment.

4.2 Reporting to the Accountable Executive

In compliance with 14 CFR § 139.402(c)(3), PDX has established a process to ensure that the AE receives regular, structured updates on the performance and implementation of the SMS through a Safety Assurance Report. These updates are essential for ensuring that the AE is informed, engaged, and able to make data-driven decisions regarding resource allocation, policy direction, and strategic support of the SMS.

The updates will include the following:

- Compliance with 14 CFR Part 139 Subpart E and Subpart D
- Performance of safety objectives
- Safety critical information distributed
- Status of ongoing mitigations
- Status of SMS implementation

4.2.1 Compliance with Subpart E and Subpart D

To ensure compliance with 14 CFR § 139.402(c)(3)(i), and to support informed executive oversight, updates to the Accountable Executive will include a summary of the airport's performance under both Subpart E (SMS requirements) and Subpart D (operational requirements) of 14 CFR Part 139.

Subpart E – Safety Management System

Compliance with Subpart E is assessed by reviewing the performance and implementation status of each core SMS component:

1. *Safety Policy:* verifies that the safety policy statement is current, actively promoted, and is accessible to all stakeholders. Confirms that roles and responsibilities are clearly defined and the SMS leadership is functioning in accordance with the manual.
2. *Safety Risk Management:* reviews whether hazards are identified, risk assessments conducted appropriately, and mitigations are documented and implemented in accordance with 14 CFR § 139.402(b)(2).
3. *Safety Assurance:* Assesses whether monitoring activities are performed and documented. Also includes status of safety objectives and confirms they are appropriately evaluated through SPIs.
4. *Safety Promotion:* Confirms that SMS training, awareness materials, and safety-critical communications are delivered to appropriate audiences.

5. *Manual Currency:* Documents whether the SMS Manual has been reviewed and updated as required, and whether any changes have been submitted to the FAA.

Subpart D – Operational Requirements

To support the AE's understanding of broader airport compliance, the report will summarize data related to Subpart D compliance, including areas that may benefit from additional attention or improvement.

4.2.2 Performance of Safety Objectives

Progress toward SMS safety objectives, as established under 14 CFR § 139.402(a)(6), is tracked using defined SPIs. The Safety Assurance Report includes data visualizations, trend summaries, and narrative explanations that illustrate whether PDX is on track to meet its safety objectives, along with recommendations for action if targets are not being achieved.

4.2.3 Distribution of Safety-Critical Information

The Safety Assurance Report confirms that all safety-critical information has been disseminated to appropriate stakeholders per 14 CFR § 139.402(d)(5)(ii). A summary of key safety communications, lessons learned, and promotional materials is included to demonstrate ongoing outreach and engagement with AOA stakeholders.

4.2.4 Status of Ongoing Mitigations

The Safety Assurance Report includes the status of all open or ongoing mitigations implemented through the PDX SRM process per 14 CFR § 139.402 (b)(2)(v). This includes implementation progress, verification status, effectiveness review outcomes, and any pending residual risk approvals. Delays or ineffective mitigations are flagged for review and possible escalation.

4.2.5 SMS Implementation Progress

The Safety Assurance Report provides a status update on the implementation of the SMS as outlined in the PDX SMS Implementation Plan. This includes completion of milestones, personnel training, development of procedures, documentation updates, and coordination with internal and external stakeholders.

Section 5 - Safety Promotion

5.1 Introduction

Safety Promotion ensures all personnel who work in the AOA are aware of safety expectations, understand their role in the SMS, know how to access the hazard reporting system, and have access to the information and training they need to make safe, informed decisions.

The goal of Safety Promotion is to foster a culture of trust, accountability, and shared responsibility. This is achieved by ensuring safety-critical information is communicated effectively, training is meaningful and accessible, and stakeholders feel empowered to report hazards and contribute to continual safety improvements.

5.2 Training

Personnel with unescorted access to the AOA will receive SMS training appropriate to their role. This includes a safety awareness orientation and specialized SMS training for individuals with operational roles under the SMS. This training must be completed at least every twenty-four months,

Safety Awareness Orientation

All personnel with access to the AOA or who perform Part 139 regulated duties must complete the Safety Awareness Orientation. This training includes:

- An overview of the PDX SMS purpose and structure
- How to identify and report a hazard
- How reported hazards are evaluated and addressed
- How feedback is provided through the SMS
- How PDX communicates important safety information

Training is delivered through the PDX Security Badging Office Computer-Based Training (CBT) and must be completed prior to gaining unescorted access to the AOA. Training is reviewed and updated by the SMS Team every twenty-four calendar months per 14 CFR § 139.402(d)(1).

Operational and Functional Training

Port of Portland personnel with responsibilities in regulated operational areas will receive additional training that includes:

- Participation in the risk assessment process
- Use of the hazard reporting system
- Integration of SMS procedures into operational duties

Training may be delivered in department-led briefings, a digital learning management system, or coordinated by the SMS Team in collaboration with department leads.

SMS Team and Supervisor / Manager Training

The SMS Team, supervisors, and mid-level managers may receive supplemental SMS training focused on:

- Leading and documenting corrective actions or mitigations
- Promoting safety reporting within their teams
- Reviewing data for trend identification
- Communicating safety expectations and lessons learned
- Participating in preliminary hazard assessments or technical risk assessments

Executive and Executive Committee-Level Training

Senior leaders who serve on the SMS Executive Committee, who have a specific role in the SMS organization chart, may receive targeted training on:

- The regulatory structure and oversight responsibilities
- Reviewing risk assessments and mitigation plans
- Escalation criteria and decision thresholds
- Understanding SMS performance data and safety objectives

5.3 Safety Communications

Safety-critical information is communicated regularly through multiple channels to ensure it reaches all relevant personnel. These communications may include:

Safety Bulletins: Issued following relevant incidents, findings, or trend analyses, to share lessons learned, or to reinforce policies or procedures, and to provide awareness of new policies or procedures.

Operational Briefings: Held by supervisors or the SMS Team to relay SMS updates or reinforce safety expectations during shift handovers, safety briefings, or other venues.

Committee Summaries: Key findings or mitigation updates from the SMS Team or SMS Airside Safety Committee are shared with stakeholders to increase transparency and reinforce engagement.

Digital Updates: The SMS web page is updated with key program documents, resources, the Safety Policy Statement, and relevant announcements.

Stakeholder Engagement: The SMS encourages two-way communication by actively seeking input from AOA stakeholders through:

- Hazard reporting and anonymous feedback tools
- Post-mitigation feedback requests
- Focused outreach to tenants, service providers, and other stakeholders
- Safety surveys or informal check-ins during inspections and audits

Participation is used to inform future communication and training strategies and to foster proactive safety culture.

Appendices

The following appendices provide additional details and tools to support the SMS

Appendix A: Abbreviations and Definitions

Acronyms and Abbreviations

AE – Accountable Executive

AOA – Air Operations Area

ARFF – Aircraft Rescue and Firefighting

ATC – Air Traffic Control

BI – Business Intelligence

CAP – Corrective Action Plan

FOD – Foreign Object Debris

eGSE – Electric Ground Service Equipment

GSE – Ground Service Equipment

LRA – Limited Risk Assessment

PDX – Portland International Airport

PHA – Preliminary Hazard Assessment

PPE – Personal Protective Equipment

RE – Responsible Executive

RM – Responsible Manager

RRT – SMS Rapid Review Team

RWY – Runway

SA – Safety Assurance

SMART (objectives) – Specific, measurable, achievable, relevant, and time-bound

SME – Subject Matter Expert

SMS – Safety Management System

SOP – Standard Operating Procedure

SPI – Safety Performance Indicator

SRM – Safety Risk Management

TRA – Technical Risk Assessment

VSR – Vehicle Service Road

Definitions

Accident – an unplanned event or series of events that results in death, injury, or damage to, or loss of, equipment or property.

Accountable Executive (AE) – an individual designated by the airport to act on its behalf for the implementation and maintenance of the SMS. The AE has control of the airport's human and financial resources for operations conducted under the 14 CFR Part 139 Airport Operating Certificate. The accountable executive has ultimate responsibility to the FAA, on behalf of the airport, for the safety performance of operations conducted under the airport's Operating Certificate.

Corrective Action Plan (CAP) – A documented strategy for eliminating or mitigating identified hazards, addressing noncompliance, or resolving deficiencies discovered through assurance or investigation processes.

Hazard – Any condition, activity, or situation that has the potential to cause injury, illness, death, damage to or loss of a system, equipment, or property. A hazard is a prerequisite to an accident or incident.

Incident – An occurrence, other than an accident, that affects or could affect the safety of operations.

Just Safety Culture – A key component of safety culture in which personnel are encouraged to report safety issues and errors without fear of reprisal.

Likelihood – The estimated frequency or probability of a specific consequence or outcome occurring due to a hazard.

Mitigation – An action or control implemented to reduce a hazard's risk.

Movement Area - the runways, taxiways, and other areas of an airport that are used for taxiing, takeoff, and landing of aircraft, exclusive of loading ramps and aircraft parking areas.

Non-Movement Area - Ramp areas, Aprons, and other areas not under positive control of the Air Traffic Control Tower.

Residual Risk – The level of risk remaining after all known and applicable mitigations have been applied.

Responsible Executive (RE) – The individual that provides oversight of safety initiatives, monitors system performance, and works with the AE to implement strategic improvements.

Risk – exposure to harm or danger.

Risk Acceptance – The certification by the appropriate leader that he or she acknowledges and accepts the safety risk that is expected.

Risk Level – The combination of the predicted severity and likelihood of the worst credible outcome of a hazard.

Risk Matrix – A tool used to assign a risk level (e.g., Low, Medium, High, Critical) by evaluating the combination of severity and likelihood.

Runway Excursion - A runway excursion is a veer off or overrun from the runway surface by an aircraft.

Runway Incursion – Any occurrence at an airport involving the incorrect presence of an aircraft, vehicle, or person on a runway.

Safety Management System – An integrated collection of processes and procedures that formalizes a proactive approach to system safety through risk management.

Safety Performance Indicator (SPI) – a measurable value used to track and assess the effectiveness of SMS. These indicators help identify trends, monitor progress, and guide decision-making.

Safety Promotion – Focused efforts to foster a positive safety culture through training, communication, and awareness.

Safety Risk Management (SRM) – A formal processes used to identify hazards, assess risk, and implement and track risk controls.

Severity – The potential impact or consequence of a hazard, typically measured by its effect on people, equipment, operations, or the environment.

SMS Airside Safety Committee – A cross-organizational committee that reviews PHAs, reports on airside safety concerns, and collaborates on strategic approaches to risk-informed decision-making.

SMS Rapid Review Team – A designated group of trained individuals responsible for conducting preliminary evaluations of reported hazards. This team determines whether further risk assessment is needed and ensures proper documentation and routing of hazards through the SRM process.

Tenant - any person or business leasing or renting space at the Airport.

Technical Risk Assessment (TRA) – A documented analysis that evaluates the risk associated with identified hazards and recommends appropriate mitigations.

Vehicle Service Road (VSR) - a designated roadway for vehicles in a Non-Movement Area.

Worst Credible Outcome – The most severe consequence of a hazard that is still consider plausible given the system design, operational environment, and existing controls.

Appendix B: PDX SMS Safety Policy Statement

Port of Portland

Portland International Airport
SMS Safety Policy Statement

At Portland International Airport (PDX), safety is a core value that guides every decision we make. Through our commitment to safety, we seek to eliminate harm by fostering a culture of active prevention, curiosity, evaluation, and action. The PDX Safety Management System (SMS) is one way that we deliver on this commitment. The SMS provides a structured approach to identifying, quantifying, and controling hazards, helping to preserve a safe environment for PDX employees, tenants, stakeholders, and the traveling public.

To uphold our commitment to safety, we will:

- Adopt and promote airport SMS at the leadership level and throughout the organization, setting clear expectations that airport employees and stakeholders prioritize and contribute to the SMS.
- Continue to engage in proactive safety risk management, data-driven decision-making, and safety assurance processes to enhance safety and continually improve.
- Encourage all airport stakeholders to report safety issues confidentially and without fear of reprisal. We will recognize and reward proactive safety behavior and foster open and transparent safety communications.
- Allocate the staff, training, technology, and financial resources needed to support safe operations at PDX.
- Comply with all applicable safety regulations and ensure that our SMS meets or exceeds federal requirements and industry standards for safe airport operations.

Trust and transparency drive safety – when issues arise, we look beyond individual actions to find and correct gaps in the system. Our leadership team is fully committed to ensuring SMS is integrated into every aspect of airside operations, and we expect the same dedication at all levels of management. We ask every employee, tenant, and partner to be vigilant, proactive, and committed to upholding the highest safety standards. Together, we will lead the way in airport safety and operational excellence.

Dan Pippenger
Chief Aviation Officer
SMS Accountable Executive

Kama T. Simonds

Stephen Nagy
Director, Airport Operations
SMS Responsible Executive

Kama Simonds
Sr. Mgr., Airside Operations & GA Airports
SMS Responsible Manager

Move with purpose.
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portofportland.com

Original Date:________________________________

Revision Date:________________________________

Appendix C: PDX SMS Area of Oversight

Port of Portland

Preparer: P. Ebert
Manager: P. Ebert
Date: May 2025
Geographic Data Standards:
Projected Coordinate System Name:
NAD 1983 HARN State Plane, Oregon North
Map Projection Name: Lambert Conformal Conic
Units: International Feet

PORTLAND INTERNATIONAL AIRPORT

Safety Management System
AOA Map

Reviewed by: Pat Ebert | MD PDX 2025-3124

R:\Maintainables\PDX\3000 - Facility Map\PDX_3000.mxd

Appendix D: PDX SMS Risk Matrix

Severity Rating	Likelihood				
	A **Rare** *(Less than once per decade)*	B **Unlikely** *(Occurs once every 5-10 years)*	C **Possible** *(Once/year or a few times every 2-5 yrs)*	D **Likely** *(Once/month or several times/yr)*	E **Frequent** *(At least once/week)*
5 Catastrophic	5A	5B	5C	5D	5E
4 Major	4A	4B	4C	4D	4E
3 Serious	3A	3B	3C	3D	3E
2 Moderate	2A	2B	2C	2D	2E
1 Minor	1A	1B	1C	1D	1E
No SMS Risk					

Original Date:___________________________________

Revision Date:__________________________________

Oversight and Decision-Making

Risk acceptance authority is tiered to escalating levels of approval that align leadership to direct and allocate resources to support hazards and applicable mitigations. No hazard may be closed or marked as "accepted" without approval from the appropriate level of authority. All risks require documentation, tracking, and monitoring.

Only Port of Portland personnel have authority in these safety decisions. While external partners (airlines, tenants, contractors, etc.) may provide risk analysis input or data, risk assessments are completed by Port staff. This section outlines who is responsible for oversight and decision-making at each risk level.

Risk acceptance authority is based on the residual risk rating and the initial risk rating should guide when leadership engagement is needed. Initial notification to SMS leadership will occur based on the initial risk rating. Notification will be provided to the designated leadership level for the assigned initial risk rating, as well as subordinate levels. See Table 3.

PDX SMS Risk Acceptance Authority				
Initial Risk Rating	**Leadership Notification**	**Residual Risk Rating**	**Acceptance Authority**	**Risk Tolerance**
Low	Responsible Manager	**Low**	RM	Acceptable with monitoring and documentation, mitigations encouraged, if practical.
Medium	Responsible Manager	**Medium**	RE	Acceptable with monitoring and documentation, mitigations preferred.
High	Responsible Executive	**High**	AE	Acceptable with documentation and tracking required. Mitigations expected.
Critical	Accountable Executive	**Critical (Unacceptable)**	Not acceptable. Interim risk acceptance by AE.	Unacceptable risk. Immediate corrective action required to reduce risk level. Mitigation strategy, tracking, and comprehensive documentation required.

Table 3. PDX SMS Risk Acceptance Authority

Low, medium, and high risks may be accepted by the acceptance authority. For any level of risk, mitigations should be considered. Critical risks require immediate corrective action to reduce the risk level. When a mitigation strategy is presented to an acceptance authority for signature, they are approving the mitigation strategy and accepting the interim (initial) risk until the mitigations can be implemented. If the authority does not approve the mitigation strategy, they are accepting the initial risk. Signature logs and approval workflows will be utilized for all risk assessments and in accordance with the PDX SMS governance structure.

This oversight model aligns with FAA SMS guidance that low risks can be managed operationally, whereas critical risks demand executive attention. It embeds safety into all levels of management, routine issues are handled at the department level, facilitated by the SMS Team, while critical issues engage senior leadership and the Accountable Executive.